



PLANET 2

WASP-76 B

- Description:** Inflated gas giant with iron rain in its clouds. It is tidally locked, and its dayside is so hot that iron evaporates, drifts to the cooler nightside and falls as rain.
- Host star:** WASP-76, an F-type star that is larger, hotter and brighter than our Sun.
- System:** The only planet known around WASP-76 so far.
- Location:** Constellation Pisces, about 634 light-years from Earth. Far too faint for the naked eye (magnitude 9.5); a telescope is needed.
- Discovery:** In 2016 by the SuperWASP survey, led by Richard West et al.

Datasheet

EXOPLANET DATA		
	TEMPERATURE	~2000 °C
	PRESSURE	No surface
	ATMOSPHERE	H ₂ , He, Fe
	GRAVITY	0.66 g
	PLANET RADIUS	
	ORBITAL PERIOD	
	SEMI-MAJOR AXIS	
	TRANSIT DEPTH	
	TRANSIT DURATION	

HOST STAR DATA		
	NAME	WASP-76
	SPECTRAL TYPE	F7V
	TEMPERATURE	6329 K
	DISTANCE	634 ly
	MASS	1.458 M _☉
	STAR RADIUS	1.756 R _☉
	AGE	1.8 Gyr
	APPARENT MAGNITUDE	9.52
	NUMBER OF PLANETS	1

SURVIVAL CHALLENGE

Aerial life in gas-giant clouds with huge day/night swings

NOTES



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NOTES

PHYSICS

 PLANET RADIUS	
 AI-ASSISTANCE	
 REFLECTIONS	

MATHEMATICS

 ORBITAL PERIOD		 SEMI-MAJOR AXIS	
 AI-ASSISTANCE			
 REFLECTIONS			

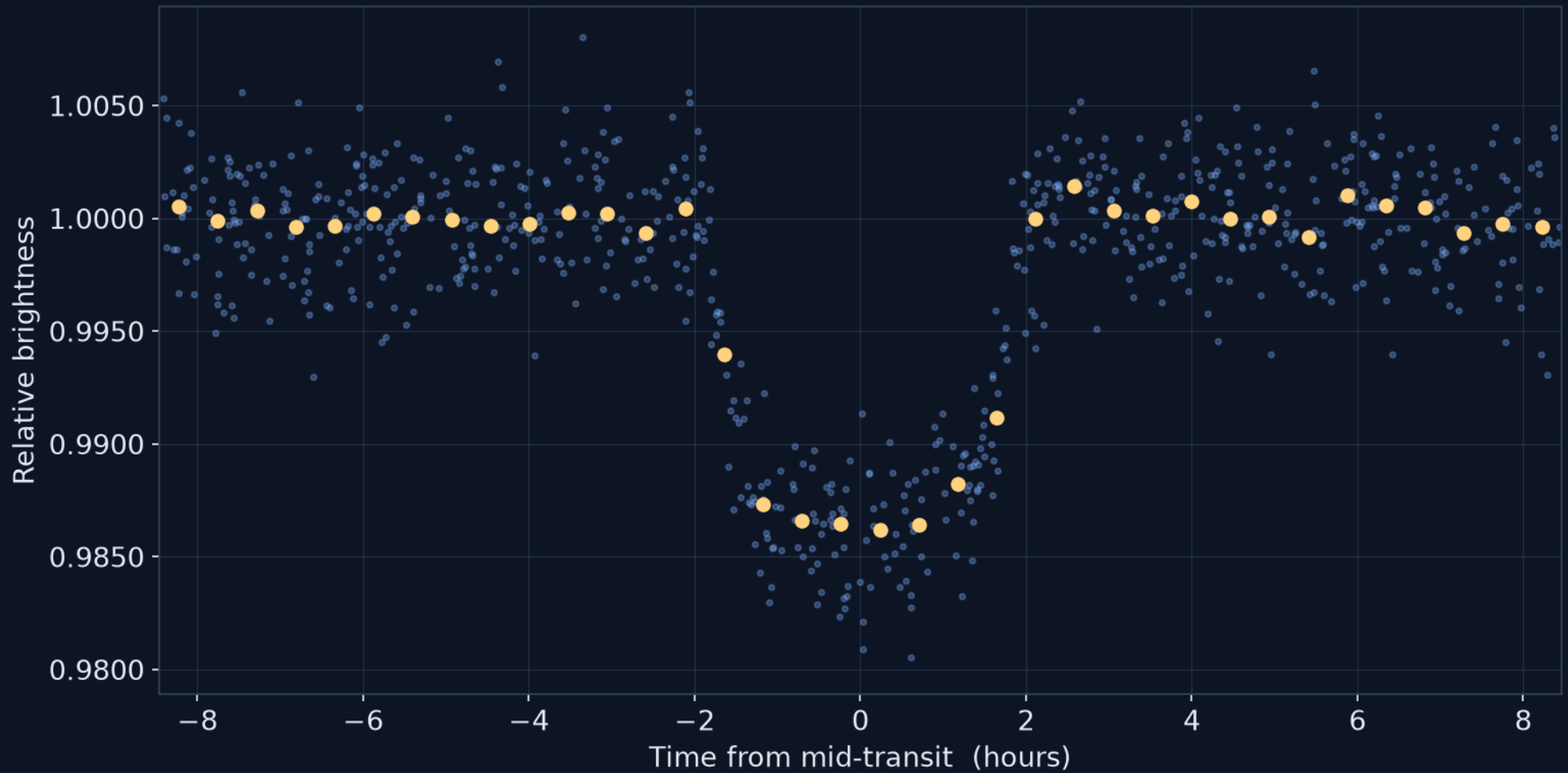
CHEMISTRY

 SAMPLE DATA	
 AI-ASSISTANCE	
 REFLECTIONS	

BIOLOGY

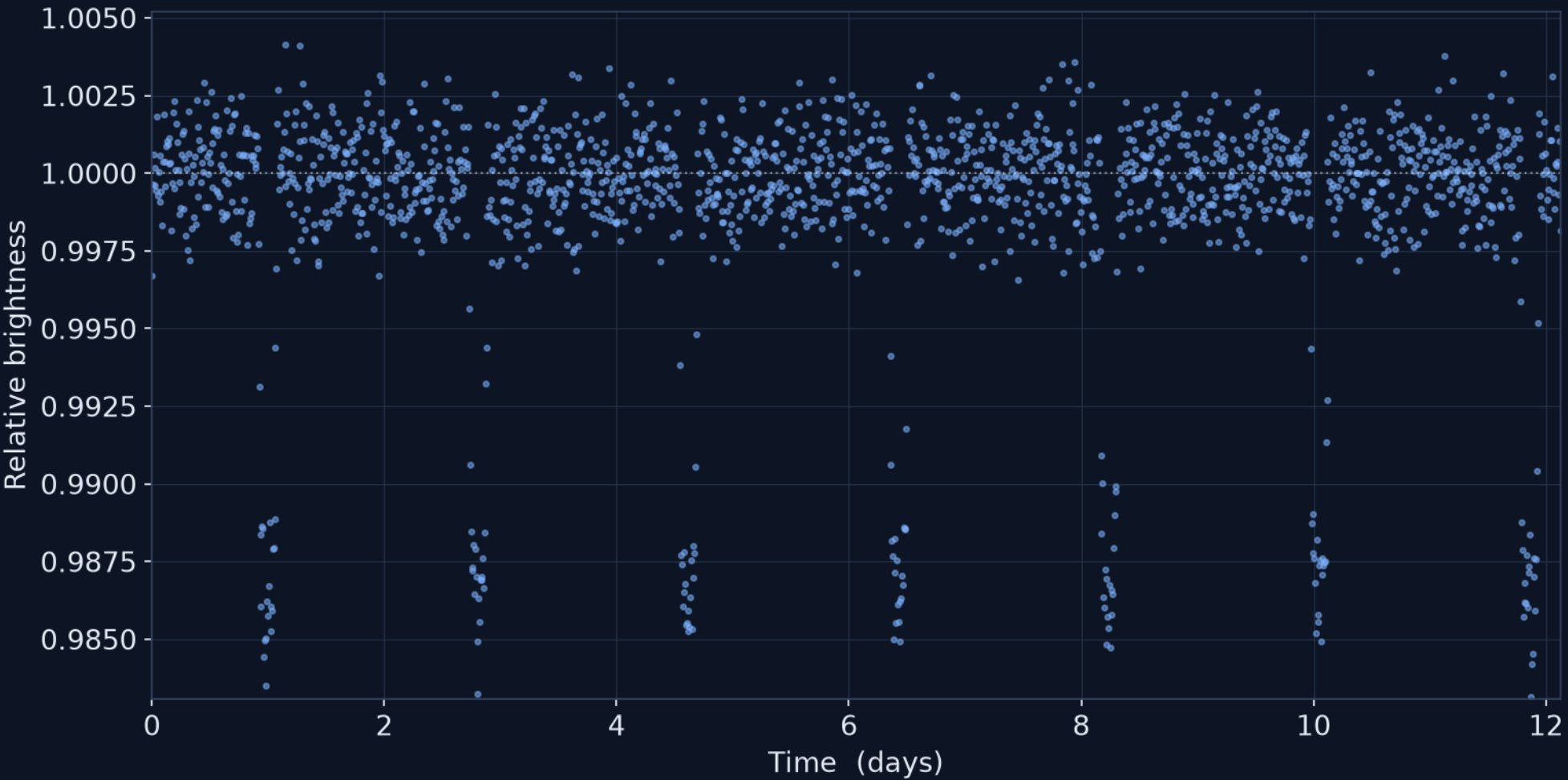
 ORGANISM	
 AI-ASSISTANCE	
 REFLECTIONS	

WASP-76 b: single transit

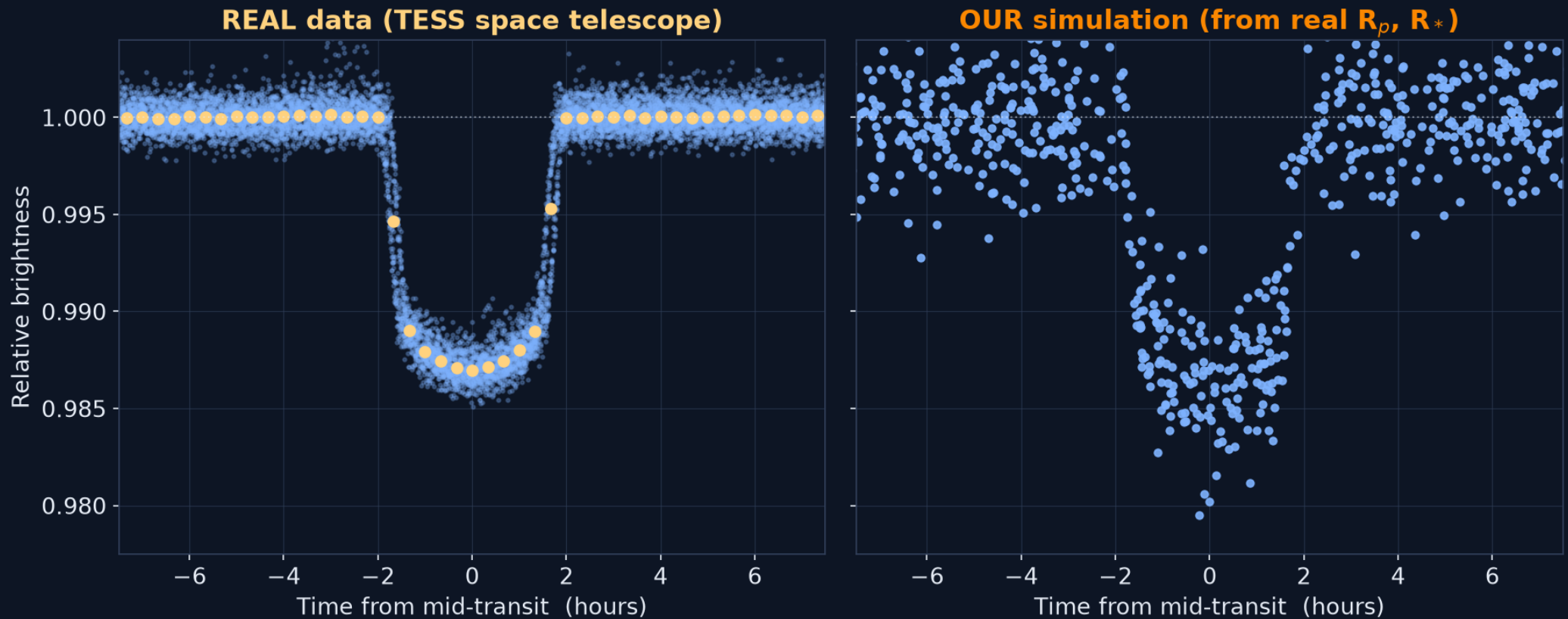


Blue points are simulated brightness measurements (many orbits phase-folded together, like real data); the gold points are binned averages.

WASP-76 b: light curve over several orbits

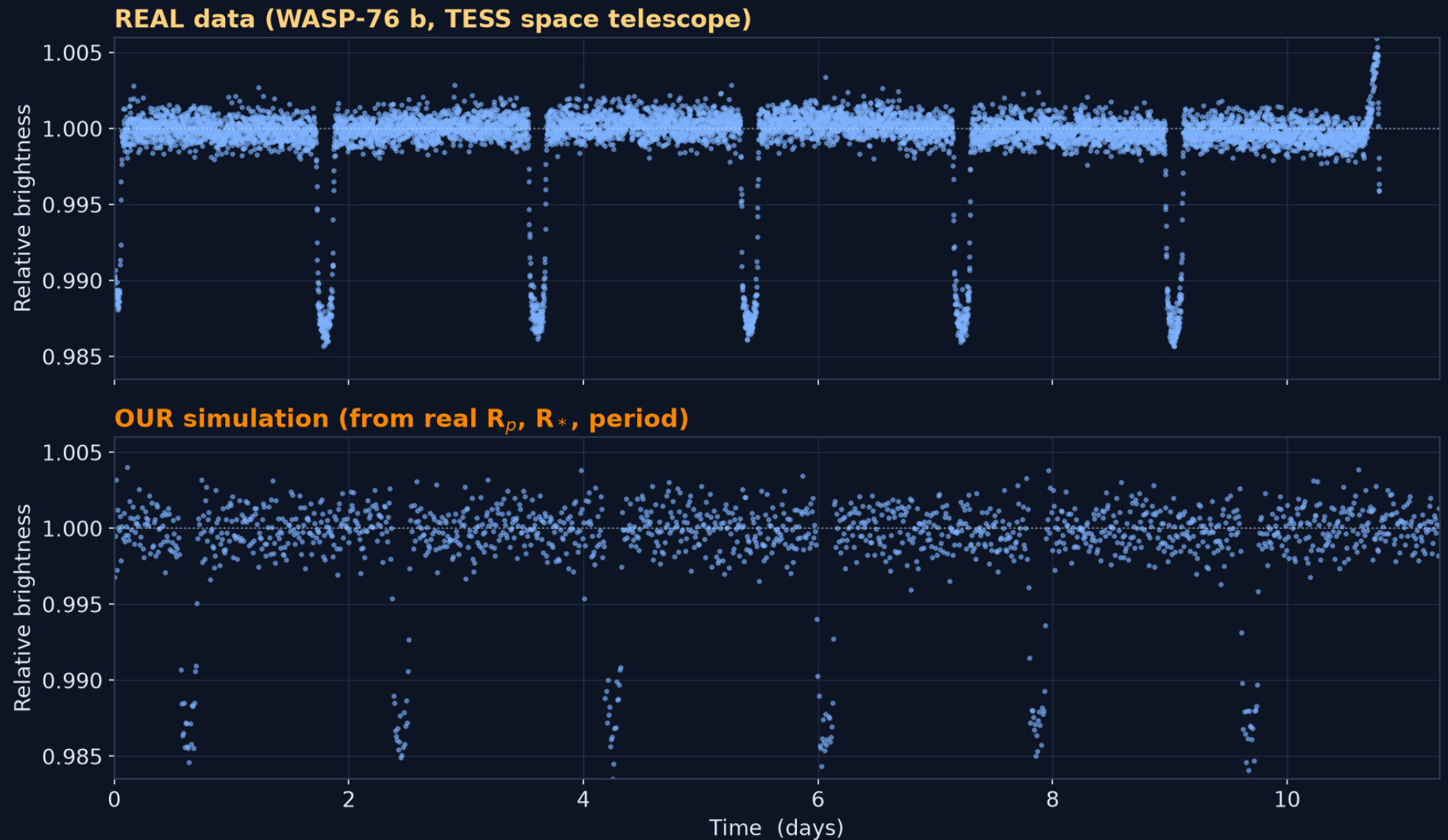


WASP-76 b: real telescope data vs. our simulation (single transit)



Left: a real transit of WASP-76 b measured by the TESS space telescope (folded over many orbits; gold points are binned averages). WASP-76 b is also a target of ESA's CHEOPS mission. Right: our simulation, built only from the planet's real radius, star radius and period. It reproduces the real numbers: same depth ($\sim 1.1\%$) and duration (~ 3.9 h). The real dip is rounded (stellar limb darkening); our model shows the same shape. Real data: TESS via MAST; points in `Transit_dataWASP76b_REAL_TESS_transit.csv`.

WASP-76 b: real vs. our simulation (light curve over many orbits)



Top: WASP-76 b's real TESS light curve over 11 days, showing repeated transits (the spacing between them is the orbital period, ~ 1.8 days). The small gap and the upturn at the end are real instrument effects. Bottom: our simulation, from the planet's real radius, star radius and period, showing the same periodic dips. Real points in `Transit_data\WASP76b_REAL_TESS_periodic.csv`.